Bhumi Lipi: Concept Note

LandTech Innovation Hackathon 2026 · Track: Digital Land Records, Verification & Digitization

Selected problem	**P2: Deed & Ownership Verification**, with P3 (fraud and double-sale), P4 (registration to e-Mutation) and P5 (citizen check) as supporting features
One-line pitch	Every land deed gets a tamper-evident digital seal, so anyone can verify ownership in 10 seconds and the same land cannot be sold twice.
Status	Working prototype, synthetic data only. Public repo with README, 51 automated tests and a browser end-to-end test. Live link and 3-minute demo video submitted with this note.

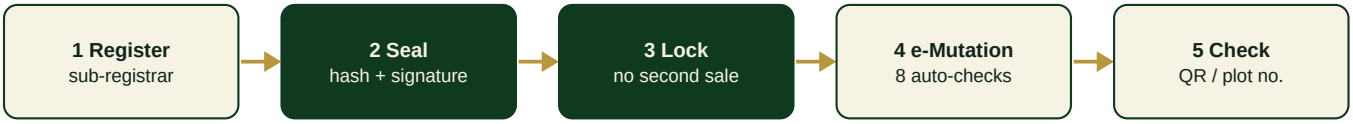
1. The problem and why it matters in Bangladesh

Land disputes are among the most common civil cases in Bangladesh, and many begin with the same few failures:

- **Forged or altered deeds.** A paper copy is easy to fake and hard to check.
- **The same land sold twice.** Registration happens at the sub-registry office; mutation happens later at the AC Land office. In between, nothing stops a second sale or a second mortgage.
- **Records that do not match.** Khatians from different survey generations (CS, SA, RS, BS) disagree, and offices do not share data.
- **Buyers cannot check.** A citizen has no cheap, trusted way to learn who owns a plot, whether a sale is pending, or whether it is government land.

We chose **P2** because verification is the foundation: when a deed can be proven genuine, altered or outdated in seconds, forgery and double-selling lose their main tool. P3, P4 and P5 then become small, natural layers on top of the same sealed records.

2. Proposed solution



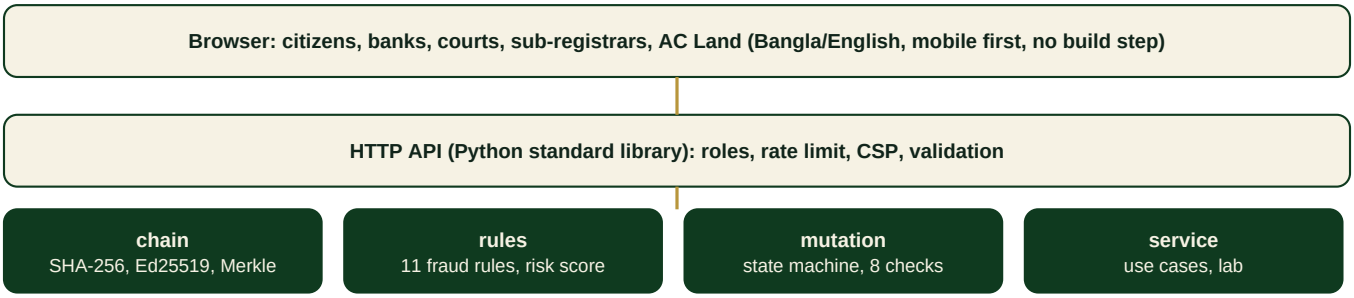
Registration and mutation become one signed, locked, auditable lifecycle. A public Merkle anchor protects history.

- **Seal.** Each deed becomes a block: a SHA-256 hash of all its fields and of the previous block of the same plot, signed with that office's own Ed25519 key. Changing any old record breaks every later hash. The deed document's hash is inside the block, so a changed file is detected.
- **Lock.** After a sale the plot locks automatically. A second sale, a fake seller, a mortgaged plot or government (khas / acquired) land is blocked, with the reason in Bangla and English.
- **e-Mutation.** Mutation starts by itself with a tracking ID and eight automatic checks (signature, owner, lien, court case, survey match, price, rapid resale, government land). The AC Land officer decides in one click. A rejection appends a REVERSAL block, so history is never erased.
- **Check.** Citizens, banks and courts see a Green / Amber / Red verdict in plain words, scan the deed QR, or drop the deed file to see if it was altered.
- **Anchor.** A public Merkle root over all records, a browser-side inclusion proof, and an independent verifier script mean even an insider with a key cannot rewrite history unnoticed.

What we learned from other countries, and what we changed

Country	Idea	Gap	Bhumi Lipi
Georgia	Deeds on a public blockchain	Needs blockchain infrastructure; no fraud rules	Hash chain + office signatures + fraud rules + public anchor. No coin, no fees
India	Drone maps and GIS titles	Maps alone do not stop forged sales	One plot ID that deeds and CS/SA/RS/BS records attach to
Rwanda	Nationwide GIS demarcation	Overlaps found late	Survey-area mismatch flagged at registration
Estonia	X-Road data layer, digital ID	Needs national ID and large integration	Registry and AC Land in one lifecycle, role-based; works with demo IDs
Zambia	Blockchain + mobile GPS	Low access in remote areas	Bangla, low-data, mobile-first check page

3. System architecture and tech stack



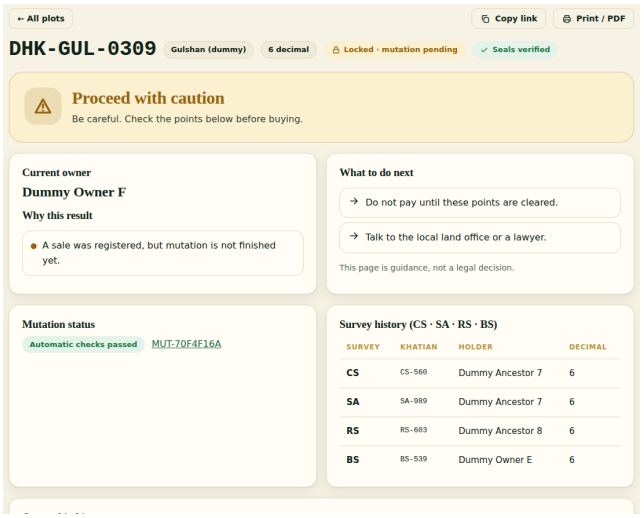
SQLite (PostgreSQL later): append-only triggers + chain guard, one signing key per office, daily Merkle anchor, independent verifier script

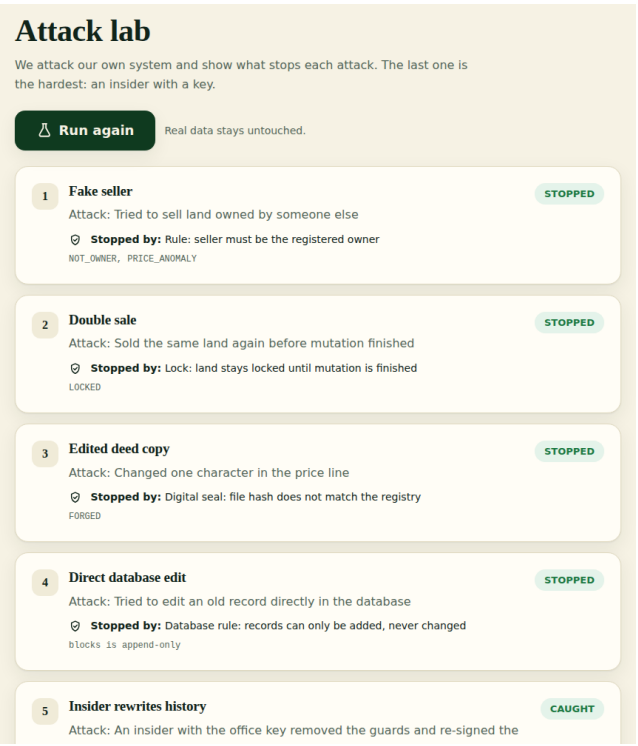
Layer	Choice	Why
Crypto	SHA-256, Ed25519 (the cryptography library), Merkle tree	Standard and audited; no home-made crypto
Backend	Python 3.12 standard library HTTP server, SQLite	One dependency, small attack surface, fully explainable; service layer is framework-free (FastAPI/PostgreSQL later)
Frontend	Vanilla JavaScript modules, own QR encoder, no build step	Fast on slow networks, easy to deploy, text inserted with textContent (no XSS)
Security	Strict CSP, input validation, rate limit, append-only DB triggers, hashed IPs	Defense in depth; threat model in docs/THREAT_MODEL.md
Quality	51 unit/integration tests, real-browser test, QR decoded with OpenCV	Every demo claim is tested

Honest limit. A single-operator chain is tamper-*evident*, not tamper-proof. Rewrites are caught by the chain check, the signatures and the public anchor. Production adds hardware-held office keys and several offices acting as witnesses.

4. Main features (all working in the prototype)

Feature	Problem
Signed hash-chain registry with append-only database rules and a fork guard	P2
Verify by QR, seal code, deed file, or "try to cheat" editor: Genuine / Outdated / Forged	P2
11 fraud rules, explainable 0-100 risk score, fake-seller and double-sale blocks, alerts queue	P3
Plot lock, e-Mutation state machine, tracking ID, officer queue, reversal blocks	P4
Citizen verdict, ownership and survey (CS/SA/RS/BS) history, print to PDF, Bangla and English	P5
Bank liens and double-mortgage block	P3
Public Merkle anchor, in-browser inclusion proof, independent verifier script	P2
Attack lab: five attacks run on a private copy, showing what stops or catches each	All





Left: citizen check page (Amber: sale registered, mutation pending). Right: attack lab results. Synthetic data.

5. Nationwide applicability

Who	What they do in Bhumi Lipi
Sub-registry offices	Register and sign deeds; see the risk before saving
AC Land / Union land offices	Receive an auto-checked mutation queue and decide in one click
Banks	Verify collateral; place and release liens; double mortgage is blocked
Land acquisition offices	Identify the verified owner before paying compensation
Survey and records offices	Attach CS/SA/RS/BS records to one plot ID (OCR for old khatians is on the roadmap)
Courts and lawyers	Ownership history with proof they can verify independently
Citizens and buyers	Check before buying; track mutation on a phone
Real estate developers	Clear-title check before purchase

Rollout. (1) Pilot in one upazila: one sub-registry office and its AC Land office. (2) District rollout with bank integration. (3) Link with existing e-registration and e-mutation systems through an API gateway. (4) National scale with hardware-held keys per office and multi-office witness nodes publishing the daily anchor. **Feasibility.** Stateless service layer that scales horizontally, PostgreSQL for production, low-data mobile pages, open-source licence, and cost dominated by hosting and key management. It works alongside existing systems instead of replacing them.

6. Team, mentors and compliance

Role	Name	DIU ID / organisation
Student member (Team Leader to be marked)	Mohammad Abrar Hasnat	253-15-689
Student member	[fill in]	[fill in]
Faculty mentor	[assigned by organizers]	[fill in]
Industry mentor	[assigned by organizers]	[fill in]

Data and rules. Synthetic data only: no real names, NID numbers, phone numbers or land records; no scraping; no testing of any live system. **AI tools:** Claude (Anthropic) helped draft code, tests and documents; both student members reviewed everything and can explain it. Code is open source (MIT). Repository, README, live link and demo video are submitted with this note.